Newton’s Laws: Measured with a Sensor Cart

U3 / S3.1

PREP — BEFORE DAY 1 AND DAY 2

WHEN	TASK	DETAIL
Days before	Charge and test	Charge the 5 sensor carts in use plus the 3 spares, and the 5 separate Go Direct force sensors (8 working carts, 5 force sensors, 5 groups: confirmed by Matt, 2026-11-12). Connect each cart to a student device once. Number carts and sensors 1–5 so groups don’t grab a neighbour’s cart in the connection list.
Days before	Check the signs	On one cart, pull the hook and push the cart forward. Note which sign the force reading gives for a pull and which sign the wheel-derived acceleration gives. If they disagree, Part B slopes come out negative. Decide in advance how you will flip one sign: the software’s sign-reverse option, or multiply one column by –1. **No menu path is written anywhere in this lab. Check it yourself.**
Before Day 1	Tracks and trays	5 tracks flat on benches, an end stop on each end. Tray per group: cart, one mass bar, 40 cm of string. Balances where every group can reach one. **Run one tow trial yourself:** if the tow reading is lost in the sensor’s noise, tell groups to use m·|a| from the coasting slope as their friction.
Before Day 2	Second sensor	Add one force sensor per group with its hook and bumper. No string: the hooks connect directly. Run the Part C test tug on your own pair. Both read a pull with the same sign until the separate sensor is reversed. Check that the two hooks actually link.

Materials — total for 5 groups

Equipment list confirmed by Matt, 2026-11-12: Go Direct Sensor Cart × 5 in use (10 owned, 2 broken, so 8 working and 3 spare) · Dynamics Track × 5 · end stops × 10 · cart mass bar × 5 · Go Direct force sensor × 5 (Day 2) · spring bumpers · balance · track pulley and hanging masses for the extension. Also: string, 40 cm × 5 · student devices with Graphical Analysis × 5. One lab across two days, also confirmed by Matt, 2026-11-12. **Balance resolution not confirmed:** the handout says “every digit it shows.”

TIMING — TWO 50-MINUTE PERIODS

MIN	PHASE	WATCH FOR
D1 0–5	Pre-lab check, trays out, connect carts	Groups connected to the wrong cart number.
D1 5–25	Part A: level, 3 push trials, 3 tow trials	Fit region that includes the push spike. Tow reading lost in noise.
D1 25–45	Part B: cart alone, then cart + bar	Negative slope, which means a sign mismatch. Pulls that run into the end stop.
D1 45–50	Cleanup	Strings off, carts off, mass bars back.
D2 0–5	Connect both sensors, zero both, test tug	Separate sensor not reversed before data collection.
D2 5–25	Part C: trials C1–C3 and the C1 stretch	Fingers on the bumper face during C3.
D2 25–45	Post-lab analysis in notebooks	Signs and units on every answer.
D2 45–50	Cleanup	

All five groups run Part C at once with their own cart and force sensor. There is no rotation and no waiting on Day 2. Post-lab work not finished in class is completed for homework. **PROVISIONAL: Matt’s call.**

FAILURE POINTS

- **Force sensor not zeroed.** Zero with nothing pulling on the hook before every run. A leftover offset shifts the Part B intercept and wrecks the Part A tow force.

- **Track not level.** A tilt adds a slope component of gravity that swamps friction in Part A. Students check a cart at rest, then compare a gentle push each way.
- **Sign convention.** If the cart reports a pull and a forward acceleration with opposite signs, the F–a slope is negative. In Part C, two sensors hooked face to face both report a pull with the *same* sign, because each measures the force on its own hook along its own axis. That is expected. Reversing the separate sensor puts both in track coordinates. After that, equal size at every instant with opposite sign is the third law.
- **Tow force in the noise.** About 0.03 N is close to the sensor's noise. If a group's tow readings scatter around zero, they use m·|a| from the coasting slope as friction (post-lab 2, 4 and 6 allow this).
- **Fit region.** In Part A students fit across the push and the catch, and the slope is meaningless. Only the coasting stretch counts.
- **String at an angle.** A string pulled up and away puts part of the pull vertical. Keep it level with the track.
- **Force sensor range.** The ±10 N / ±50 N range switch belongs to the older Vernier Dual-Range Force Sensor. Your sensor model is not confirmed. If yours have no switch, ignore this. If they do, ±10 N suits this lab.

Alternatives and backups

Two sensor carts colliding. With 8 working carts and 5 in use, the 3 spares make one extra pair. Two carts meeting bumper to bumper on one track give the same mirror-image force graphs, and both carts record acceleration, so $m_1a_1 \approx m_2a_2$ can be checked. Good as a demo or for a group that finishes early. **A separate force sensor standing in for a cart.** A force sensor mounted on a plain cart can stand in for one sensor cart's force reading. If your separate sensors are the Force and Acceleration model, they also record acceleration. If not, the m·a check only works for the sensor cart.

EXPECTED RESULTS — SAMPLE DATA

These numbers are **plausible sample data, not real measurements**. Replace the cart mass with your own balance reading. A good group gets F–a slopes within about 5% of the balance mass and peak-force differences of a few percent. A sloppy-but-passing group gets slopes within about 15% and still sees mirror-image graphs.

QUANTITY	ACCEPTED	SAMPLE	TYPICAL RANGE
Coasting a, level track	0 if no friction	−0.096 ± 0.008 m/s²	small, negative
Tow force at constant v	none	+0.03 ± 0.02 N	near the sensor's noise
F–a slope, cart alone	0.305 kg (sample balance)	0.318 kg · 4.3%	within 15%
F–a slope, cart + bar	0.802 kg (sample balance)	0.781 kg · 2.6%	within 15%
Peak-force percent difference	0%	2.8% · 3.0% · 3.3%	under about 10%

SAFETY + DISPOSAL

- No chemicals and no glassware are used, so no SDS applies. The safety bullets cover mechanical hazards only: carts rolling off the track end, pinched fingers at the bumper, and a falling hanging mass. **Not yet checked against district policy. Matt to confirm before copying.**
- **Eye protection: Matt to set.** The handout says "follow the room's standing eye-protection rule" and does not require goggles. The kept Alconox item 6 says "goggles come off last," which contradicts that until the block is resolved.
- **Alconox block: kept verbatim, items 4–6, per the locked template rule. It does not fit this lab.** There is no glassware and no goggles are called for, so items 4–6 tell students to do things that don't exist. The S1.3 exception that dropped it is recorded as not a precedent, so dropping it here needs Matt's own decision and its own record. If he says drop it, delete items 4–6 on the handout.

WORKED SAMPLE ANALYSIS — SAMPLE DATA, NOT REAL MEASUREMENTS

Parts A and B: positive is the direction of the push or pull. Part C: positive is toward the person holding the separate sensor, with that sensor reversed so both readings are in track coordinates. Sample cart 0.305 kg, sample bar 0.497 kg, so cart + bar is 0.802 kg.

Part A — first law

Push-trial slopes: −0.089, −0.104, −0.096 m/s². Tow forces: +0.02, +0.05, +0.03 N. Tow-stretch accelerations: +0.004, −0.006, +0.002 m/s².

mean $a = \frac{-0.089 + (-0.104) + (-0.096)}{3} = \frac{-0.289\ \text{m/s}^2}{3} = -0.0963\ \text{m/s}^2$

uncertainty $= \frac{-0.089 - (-0.104)}{2} = \frac{0.015\ \text{m/s}^2}{2} = 0.0075\ \text{m/s}^2$

$a = -0.096 \pm 0.008\ \text{m/s}^2$

mean tow force $= \frac{+0.02\ \text{N} + 0.05\ \text{N} + 0.03\ \text{N}}{3} = +0.033\ \text{N}$; uncertainty $= \frac{0.05\ \text{N} - 0.02\ \text{N}}{2} = 0.015\ \text{N}$

$F_{tow} = +0.03 \pm 0.02$ N, with $a = 0.000 \pm 0.005\ \text{m/s}^2$ during the tow

Coasting check: $f = m \cdot |a| = (0.305\ \text{kg})(0.096\ \text{m/s}^2) = 0.029$ N. Use this as f if the tow reading is lost in noise.

Part B — second law

Cart alone: slope 0.318 kg, intercept +0.041 N. percent error $= \frac{|0.318\ \text{kg} - 0.305\ \text{kg}|}{0.305\ \text{kg}} \times 100 = \frac{0.013\ \text{kg}}{0.305\ \text{kg}} \times 100 =$ **4.3%**

Cart + bar: slope 0.781 kg, intercept +0.046 N. percent error $= \frac{|0.781\ \text{kg} - 0.802\ \text{kg}|}{0.802\ \text{kg}} \times 100 = \frac{0.021\ \text{kg}}{0.802\ \text{kg}} \times 100 =$ **2.6%**

Slope rise = 0.781 kg − 0.318 kg = 0.463 kg, against 0.497 kg added.

Slope rise against bar mass: $\frac{|0.463\ \text{kg} - 0.497\ \text{kg}|}{0.497\ \text{kg}} \times 100 = \frac{0.034\ \text{kg}}{0.497\ \text{kg}} \times 100 = 6.8\%$ low

Part C — third law

C1 direct hook-to-hook pull: +1.84 N on cart, −1.79 N on hand sensor. Use the sizes. average $= \frac{1.84\ \text{N} + 1.79\ \text{N}}{2} = 1.815$ N

percent difference $= \frac{|1.84\ \text{N} - 1.79\ \text{N}|}{1.815\ \text{N}} \times 100 = \frac{0.05\ \text{N}}{1.815\ \text{N}} \times 100 =$ **2.8%**

C2 tugs: +2.31 N on cart and −2.38 N on hand sensor. $\frac{0.07\ \text{N}}{2.345\ \text{N}} \times 100 =$ **3.0%** C3 push: −1.47 N on cart and +1.52 N on hand sensor. $\frac{0.05\ \text{N}}{1.495\ \text{N}} \times 100 =$ **3.3%**

C1 stretch: mean F on cart +0.52 N, mean a +1.58 m/s².

$m \cdot a = (0.305\ \text{kg})(+1.58\ \text{m/s}^2) = +0.482$ N $\quad \sum F = F - f$, with f from Part A: $\sum F = +0.52\ \text{N} - 0.03\ \text{N} = +0.49$ N

average $= \frac{0.482\ \text{N} + 0.49\ \text{N}}{2} = 0.486$ N percent difference $= \frac{|0.482\ \text{N} - 0.49\ \text{N}|}{0.486\ \text{N}} \times 100 =$ **1.6%**

PRE-LAB KEY

1. Net force on a cart at a constant 0.40 m/s. $\sum F = 0$. Constant velocity means a = 0, and the first law says that only happens when the net force is zero. The push stopped acting the moment the hand left. No credit for "a small forward force keeps it going."
2. Slope of F against a. Mass. Units of slope: $\frac{\text{N}}{\text{m/s}^2} = \frac{\text{kg·m/s}^2}{\text{m/s}^2} = \text{kg}$.
3. Force of cart on sensor when the sensor pulls directly on the cart's hook with +2.0 N. −2.0 N, the same size, pointing back toward the cart. The two forces act on different objects. Only the +2.0 N acts on the cart, so the cart accelerates. Partial credit for −2.0 N with no reason.

4 Part A, step 3. Zero it, with nothing pulling on the hook. For the tow, the string hangs slack.

5 Region of v–t to fit. Only the coasting stretch after the hand leaves the cart. Leave out the push and the catch.

6 Accepted value in Part B. The balance mass. With a bar on, mass the cart and bar together on the balance. Do not add a stamped value.

7 Part C, step 2. Each sensor measures the force on its own hook along its own axis, and the two face each other, so both report a pull the same way. After one is reversed, the cart reads positive and the hand sensor negative.

8 Safety. End stops on both ends of the track. Hold each sensor by its body, with fingers off the front face and out of the gap. Eye protection per the room's standing rule.

9 Data tables drawn. Full credit = 3 tables drawn in ink, with units in the row labels.

POST-LAB KEY — FROM THE SAMPLE DATA

1 Coasting a with uncertainty. -0.096 ± 0.008 m/s². Tiny next to the 1–2 m/s² of Part B, so close to zero, but the range −0.104 to −0.088 does not include zero. Rolling friction points opposite the motion, which is the negative direction. "Human error" earns nothing.

2 m|a| against tow force. (0.305 kg)(0.096 m/s²) = 0.029 N, against $+0.03 \pm 0.02$ N. While coasting, friction is the only horizontal force, so f = m|a| (second law). While towing at constant velocity a = 0, so $\sum F = 0$ and the tow force equals friction (first law). They agree within the tow uncertainty. Full credit also for a group that says its tow reading was lost in noise and picks m|a| as the better friction, because it comes from a clean fitted slope.

3 Percent error for each slope. 4.3% and 2.6%. Full work under Part B of the worked sample analysis, page 3. Units carried in every step.

4 Slope rise and intercept. Up 0.463 kg against 0.497 kg added, 6.8% low. The intercept (+0.041 N, +0.046 N) is the force reading at a = 0, the force that only balances friction. It is comparable to the Part A friction of about 0.03 N, higher by 0.01–0.02 N, which is inside the tow uncertainty. The mass changed, so the slope changed.

5 Percent difference of peaks; signs. 2.8%, 3.0%, 3.3%. The opposite signs mean opposite directions, and the mirror-image curves show the sizes are equal at every instant, not only at the peak. The forces don't cancel because one acts on the cart and the other acts on the hand sensor. Forces cancel only when they act on the same object.

6 m·a against $\sum F = F - f$. +0.482 N against +0.49 N, 1.6% apart. The force sensor reads only the pull. Friction also acts on the cart, opposite the motion, so $\sum F = F - f$. Either Part A friction value (+0.03 N tow, or 0.029 N from m|a|) gives +0.49 N. Common wrong answer: comparing m·a with 0.52 N directly. Partial credit if the friction reason is given.

7 Unzeroed sensor reading +0.15 N. Every force reads 0.15 N high, so the intercept rises by 0.15 N and the slope does not change. Their mass and percent error are unaffected. Common wrong answer: "the mass comes out too big." No credit without the slope-against-intercept reason.

8 Track tilted downhill. Gravity now has a component along the motion. The slope becomes less negative, zero, or even positive, and it hides friction. Part A step 1: the cart-at-rest check and the gentle push each way.

9 Connection to 1.3. Part A step 4: the linear-fit slope of v–t is the acceleration. The area under v–t is displacement, in metres.

GRADING

No point split for Physics labs is set yet. Physics gradebook weights are still an open question, so Matt sets this. The carbonless copy is graded individually.

MISSED LAB

If the setup is already torn down, the student completes a Missed Lab Analysis rather than re-running the lab. Give them one group's Tables 1–3 from the same period, plus a printout or screenshot of that group's F–a graph and one Part C force overlay. They answer post-lab questions 1–8 from that data and question 9 from their 1.3 notes.